

Arun Lama

Nijgadh, Nepal | lamaarun2001@gmail.com | +977 9742432147 | [LinkedIn](#) | [GitHub](#) | [Kaggle](#) | [Tableau](#)

PROFESSIONAL SUMMARY

Computer Engineering student (graduating 2025) with 4+ years of applied experience in Python, data analysis, and machine learning. Proven ability to build end-to-end ML pipelines, engineer predictive models, and translate complex datasets into actionable insights. Hands-on with TensorFlow, PyTorch, SQL, and Tableau. Active Kaggle competitor with a track record of delivering data-driven solutions to real-world problems.

EXPERIENCE

Independent ML Researcher & Kaggle Competitor

Self-Directed | January 2021 – Present

- Entered 3–5 Kaggle competitions, progressing from data exploration to full model pipelines; achieved top ~20% finish in Walmart Sales Forecasting (115k predictions, WMAE \$1,273).
- Published notebooks and project code publicly on Kaggle and GitHub, documenting methodology, findings, and reproducible pipelines for each project.
- Built and deployed 3 end-to-end ML/AI projects independently — spanning time series forecasting, financial EDA, and a production-grade RAG system with LLM optimization.
- Self-studied advanced ML topics (deep learning, NLP, LLM deployment) through DeepLearning.AI and freeCodeCamp, applying concepts directly to project work.

PROJECTS

Amazon Stock Price Analysis

EDA | Time Series | Python

- Cleaned and analyzed 5+ years of AMZN historical data; identified price trend patterns using pandas and Matplotlib.
- Visualized volatility trends and moving averages across multiple indicators in a consolidated dashboard view.

Walmart Sales Forecasting — Kaggle Competition

Time Series Forecasting | XGBoost & LightGBM Ensemble

- Merged 4 datasets into a 421k-row corpus (45 stores, 81 depts); engineered 40+ features including lag (1/4/52 weeks), rolling stats, holiday flags, and economic interaction terms.
- XGBoost/LightGBM ensemble (60/40) with K-means dept clustering and 5× holiday weights achieved WMAE \$1,273 vs. baseline \$2,614 (**51% reduction**); ranked top ~20% on Kaggle.

Semantic Cache-Enhanced RAG System for Document Q&A

NLP | Retrieval-Augmented Generation | LLM Optimization

- Built a RAG pipeline combining FAISS vector search (all-MiniLM-L6-v2, 384-dim) with quantized Phi-2 (2.7B, 4-bit) LLM; runs on 2GB VRAM / 8GB RAM.
- Semantic cache (cosine sim ≥ 0.92) cut redundant LLM calls by 60%, reducing query latency from 2.5s to <0.4s; evaluated via BLEU-1 with Gradio chat interface.

TECHNICAL SKILLS

Languages: Python, SQL

ML / AI: Supervised & Unsupervised Learning, Neural Networks, NLP, Retrieval-Augmented Generation (RAG), Feature Engineering, Time Series Forecasting, Model Evaluation

Frameworks & Tools: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, LangChain, FAISS, Sentence-Transformers, pandas, NumPy, Gradio, Jupyter, Git

Data Visualization: Tableau, Matplotlib, Seaborn, Excel (Advanced)

Core Competencies: Data Analysis, ETL Pipelines, Statistical Modeling, EDA, LLM Optimization, Prompt Engineering

EDUCATION

Bachelor of Engineering — Computer Engineering

IOE Purwanchal Campus, Tribhuvan University | Dharan, Nepal | 2021 – 2025

CERTIFICATIONS

- **Market Research Analyst Program Powerskill** — Rochester Institute of Technology (RIT) × Excelerate (2025)
- **Advanced Learning Algorithms** — Coursera (DeepLearning.AI) (2024)
- **Supervised Machine Learning: Regression and Classification** — Coursera (DeepLearning.AI) (2023)
- **Unsupervised Learning, Recommenders, Reinforcement Learning** — Coursera (DeepLearning.AI) (2024)
- **Data Analysis with Python** — freeCodeCamp (2024)